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# 328653(28)

## B. E. (Sixth Semester) Examination, Nov.-Dec. 2015

**(New Scheme)**

**(ET & T Engg. Branch)**

## MICROCONTROLLER AND EMBEDDED SYSTEMS

***Time Allowed : Three hours***

***Maximum Marks : 80***

***Minimum Pass Marks : 28***

***Note :*** *Attempt all questions. Part (a) of each question is compulsory and carrying 2 marks each and attempt two parts from (b), (c) and (d) carrying 7 marks each.*

### Unit-I

1. (a) What is the default context of SP? Write an instruction to shift stack to scratch pad RAM. 2



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(b) Draw and explain RESET and clocking circuit in 8051. How long RESET pin should go high for successful Reset? Why RESET is referred as $6^{th}$ interrupt. 7

(c) Draw and explain in detail structure of on chip RAM. 7

(d) Ten hexadecimal bytes are residing at location starting from address 30H of internal RAM. Assumming that Hexadecimal bytes are not greater than 63H, write an ALP to convert them into packed BCD form and save them at same place after conversion. 7

### Unit-II

2. (a) Write 8 bit format and function of each bit of TMOD register of 8051. 2

(b) Write an ALP to produce square wave of frequency 3·0 kHz with 66% duty cycle on port pin P2·2. If XTAL frequency is 16 MHz use timer 0 in mode 2. 7

(c) Using timer interrupts generate to square waves at port pins P1·5 and P2·5 with frequencies 3 kHz and 10 kHz respectively. Assuming XTAL = 11·0592 MHz. 7

(d) Write 8 bit format and explain function of each bit of TCON, IE and IP registers of 8051. 7



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## Unit-III

3. (a) Write 8 bit format and function of each bit of SCON register of 8051. 2

(b) Write an ALP to receive message "GOD IS GREAT" along with null character serially at baud rate of 9600 through RXD pin. Save the characters received at address 1000H onwards in external data memory. 7

(c) What is the need of MAX232 in serial communication between embedded system and IBM PC? With proper block dingram and interfacing diagram explain the role of MAX 232 IC. 7

(d) Write short note on RS422 Bus standard. 7

## Unit-IV

4. (a) If the starting address is 4000H for interfacing external data memory 16KB ROM with 8051, determine ending address. 2

(b) Design a memory system for 8031 microcontroller using 4KB of program code ROM, 4KB of data memory ROM and 4KB of data memory RAM. 7

(c) Interface ADC 0804 with microcontroller 8051 and write suitable program to convert analog input into



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digital output and send it to external hardware through port $P_0$ continuously. 7

(d) A switch is connected to port pin P2·2. Write a program to monitor the switch continously and perform following functions : 7

(i) If SW = 0, run stepper motor clockwise

(ii) If SW = 1, run stepper motor Anticlockwise

## Unit-V

5. (a) Define an embedded systems. 2

(b) Discuss different characteristics of real time embedded systems. 7

(c) What are the basic requirements of designing an embedded system? 7

(d) Discuss different design challenges and issues in development of embedded system. 7



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